

List 3 - Fin. Ele.

1. Consider the finite dimensional Hilbert space $\mathbb{R}^n$. Provide two examples of continuous linear functionals and their Riez representatives, for different inner products.

(1) $l(v) = 0$

It's continuous, as it is limited. Representatives:

Dot product

$$(\vec{0}, v) = \sum_{i=1}^n 0 \cdot v_i = 0 = l(v) \Rightarrow \vec{0} \text{ is the representative}$$

Matrix induced inner product. Let A be symmetric, positive-definite.

$$(u, v)_A = u^T A v$$

$$(v, \vec{0})_A = v^T A \vec{0} = v^T \vec{0} = 0 \Rightarrow \vec{0} \text{ is the representative}$$

(2) $l_i(v) = v_i$

With dot product:

It's bounded:

$$|l_i(v)|^2 = |v_i|^2 \leq \sum_{j=1}^n v_j^2 = \|v\|_2^2 \Rightarrow$$

$$\Rightarrow |l_i(v)| \leq \|v\|_2$$

Finding the representative:

$$(u, v) = \sum_{j=1}^n u_j v_j = v_i, \forall v \in V \Rightarrow u_j = 0, j \neq i$$

$$u = \begin{bmatrix} u_1 \\ \vdots \\ u_i \\ \vdots \\ u_n \end{bmatrix} = \begin{bmatrix} 0 \\ \vdots \\ 1 \\ \vdots \\ 0 \end{bmatrix}$$

Matrix induced inner product:

It's bounded:

$$|l_i(v)|^2 = |v_i|^2$$

$$u^T A v = u^T (v_1 a_1 + \dots + v_n a_n) = v_1 u^T a_1 + \dots + v_n u^T a_n$$

Let $\lambda_1 > 0$ be the smallest eigenvalue of A

Using the Rayleigh quotient

$$R(v) = \frac{v^T A v}{v^T v} \geq \lambda_1 \Rightarrow v^T A v \geq \lambda_1 \|v\|_2^2 \geq \lambda_1 |v_i|^2$$

$$\Rightarrow |v_i|^2 \leq \frac{v^T A v}{\lambda_1} \Rightarrow |v_i| = |l_i(v)| \leq \frac{1}{\sqrt{\lambda_1}} \|v\|_A$$

The representative:

$$(u, v)_A = u^T A v = v_1 u^T a_1 + \dots + v_n u^T a_n = v_i, \forall v \in V \Rightarrow$$

$$\Rightarrow u^T a_j = 0, j \neq i \Rightarrow u \text{ is orthogonal to } \text{span}\{a_1, \dots, a_{i-1}, a_{i+1}, \dots, a_n\}$$

$$\Rightarrow v_i u^T a_i = v_i \Rightarrow u^T a_i = 1$$

$\therefore u$ is the vector such that $u^T a_i = 1$ and u is orthogonal to $\text{span}(\{a_1, \dots, a_n\} \setminus \{a_i\})$

2. Consider the space generated by the monomials $\{1, x, x^2, \dots, x^n\}$ defined in the interval $(-1, 1)$. Provide two examples of continuous linear functionals and their Riez representatives by defining appropriate inner products. Is this a Hilbert space? What is the Riez representative for $l(v) = v(0)$ when considering the $L^2(-1, 1)$ inner product?

$$V = \text{span}\{1, x, \dots, x^n\}$$

With L^2 inner product

$$(u, v)_{L^2} = \int_{-1}^1 u(x) v(x) dx$$

V is complete:

Note first that if $u \in V$, u is limited. Let

$$\|u\|_{L^2} = \left(\int_{-1}^1 u^2(x) dx \right)^{\frac{1}{2}} \leq \left(2 u_{\max}^2 \right)^{\frac{1}{2}} = \sqrt{2} |u_{\max}| \quad (1)$$

$u_{\max} = \sup_{x \in (-1, 1)} |u(x)|$

This implies that if u_k is a Cauchy sequence in $\|\cdot\|_{L^2}$, then u_k is a Cauchy sequence in $\|\cdot\|_{L^\infty}$.

Take $u_k(x) = a_{0k} + a_{1k}x + \dots + a_{nk}x^n$ a Cauchy sequence with $\|\cdot\|_{L^\infty}$. Take $\varepsilon > 0$. In particular, as $\{1, x, \dots, x^n\}$ forms a base, $\forall i = 1, \dots, n$, we have: $\exists M > 0$ such that $m, k > M \Rightarrow$

$|a_{ik} - a_{im}| < \varepsilon$. As $\mathbb{R}$ is complete, $a_{ik} \rightarrow a_i$. So $u_k \rightarrow u = a_0 + a_1x + \dots + a_nx^n$.

Now, because of (1), $u_k \rightarrow u$ with $\|\cdot\|_{L^2}$. So V is a Hilbert space with the L^2 inner product.

First linear functional

$l(v) = 0$. Clearly bounded

Representative: $u = 0$

In fact:

$$(0, v)_{L^2} = \int_{-1}^1 0 \cdot v(x) dx = 0$$

Second linear functional:

$$l(v) = \int_{-1}^1 v(x) dx$$

Boundedness:

By Cauchy-Schwarz inequality

$$\left(\int_{-1}^1 v(x) \cdot 1 dx \right)^2 \leq \int_{-1}^1 v^2(x) dx \cdot \int_{-1}^1 1^2 dx = 2 \int_{-1}^1 v^2(x) dx$$

$$\Rightarrow \left| \int_{-1}^1 v(x) dx \right| \leq \sqrt{2} \|v\|_{L^2}$$

Finding the representant:

$$(u, v)_L = \int_{-1}^1 u(x)v(x)dx = \int_{-1}^1 v(x)dx \Rightarrow u(x) = 1$$

$$\therefore u \equiv 1$$

$$l(v) = v(0)$$

$$(u, v)_L = l(v), \forall v \in V \Leftrightarrow$$

$$\Leftrightarrow \int_{-1}^1 u(x)v(x)dx = v(0) \Leftrightarrow$$

3. Show that a scalar product $(\cdot, \cdot)$ is a continuous bilinear form.

Let $u, v \in V$. By Cauchy-Schwarz inequality:

$$(u, v) \leq \|u\|_V \|v\|_V$$

Where $\|\cdot\|_V$ is the induced norm.

4. Complete all the details to show that a symmetric, continuous and coercive bilinear form satisfy the definition of inner product given in Definition 2.2.2. Also show that if for a sequence of functions $\{v_n\}$ in V , $\|v_n - v\|_V \xrightarrow{n \rightarrow \infty} 0$, $v \in V \Rightarrow$

$$\|v_n - v\|_a \xrightarrow{n \rightarrow \infty} 0.$$

Symmetry: follows from the symmetry of the bilinear form

Linearity: $a(\alpha v + \beta w, u) = \alpha a(v, u) + \beta a(w, u)$, by the bilinearity

Positivity: $a(u, u) \geq \alpha \|u\|_V^2 \geq 0$, $\alpha > 0$

$$a(u, u) = 0 \Leftrightarrow u = 0$$

$$(\Leftarrow) u = 0 \Rightarrow a(u, u) = a(u + u, u) = 2a(u, u) \Rightarrow a(u, u) = 0$$

$$(\Rightarrow) a(u, u) = 0. \text{ We have } a(u, u) = 0 \geq \alpha \|u\|_V^2 \Rightarrow \|u\|_V = 0 \Rightarrow u = 0$$

Let $\{v_n\} \in V$ such that

$$\|v_n - v\|_V \xrightarrow{n \rightarrow \infty} 0, \quad v \in V$$

We have, as a is continuous:

$$\|v_n - v\|_a = \sqrt{a(v_n - v, v_n - v)} \leq \sqrt{C_\alpha} \|v_n - v\|_V \xrightarrow{n \rightarrow \infty} 0$$

$$\Rightarrow \|v_n - v\|_a \xrightarrow{n \rightarrow \infty} 0$$